



Genetic risk classification for adults with AML receiving less-intensive therapies: the 2024 ELN recommendations

Hartmut Döhner,¹ Courtney D. DiNardo,² Frederick R. Appelbaum,³ Charles Craddock,⁴ Hervé Dombret,⁵ Benjamin L. Ebert,⁶ Pierre Fenaux,⁷ Lucy A. Godley,⁸ Robert P. Hasserjian,⁹ Richard A. Larson,¹⁰ Ross L. Levine,¹¹ Yasushi Miyazaki,¹² Dietger Niederwieser,¹³ Gert Ossenkoppele,¹⁴ Christoph Röllig,¹⁵ Jorge Sierra,¹⁶ Eytan M. Stein,¹¹ Martin S. Tallman,¹¹ Hwei-Fang Tien,¹⁷ Jianxiang Wang,¹⁸ Agnieszka Wierzbowska,¹⁹ Andrew H. Wei,²⁰ and Bob Löwenberg²¹

¹Department of Internal Medicine III, Ulm University Hospital, Ulm, Germany; ²Department of Leukemia, The University of Texas MD Anderson Cancer Center, Houston, TX; ³Division of Hematology and Oncology, University of Washington and Fred Hutchinson Cancer Research Center, Seattle, WA; ⁴University of Warwick Trials Unit, Centre for Clinical Haematology, Queen Elizabeth Hospital, Birmingham, United Kingdom; ⁵Department of Hematology, Saint-Louis Institute for Research, Université Paris Cité, Hôpital Saint-Louis, Assistance Publique-Hôpitaux de Paris, Paris, France; ⁶Department of Medical Oncology, Dana-Farber Cancer Institute, Boston, MA; ⁷Service d'Hématologie Séniors, Hôpital Saint-Louis, Assistance Publique des Hôpitaux de Paris and Université de Paris Cité, Paris, France; ⁸Division of Hematology/Oncology, Department of Medicine, Robert H. Lurie Comprehensive Cancer Center, Northwestern University, Chicago, IL; ⁹Department of Pathology, Massachusetts General Hospital, Boston, MA; ¹⁰Department of Medicine and Comprehensive Cancer Center, The University of Chicago, Chicago, IL; ¹¹Department of Medicine, Human Oncology and Pathogenesis Program and Leukemia Service, Memorial Sloan Kettering Cancer Center, New York, NY; ¹²Department of Hematology, Atomic Bomb Disease Institute, Nagasaki University, Nagasaki, Japan; ¹³Division of Hematology and Medical Oncology, University of Leipzig, Leipzig, Germany; ¹⁴Department of Hematology, Amsterdam University Medical Center, Location Vrije Universiteit Medical Center, Amsterdam, The Netherlands; ¹⁵Department of Internal Medicine I, University Hospital Carl Gustav Carus TU Dresden, Dresden, Germany; ¹⁶Hematology Department, Hospital Santa Creu i Sant Pau, Barcelona, Spain; ¹⁷Division of Hematology, Department of Internal Medicine, National Taiwan University Hospital, Taipei, Taiwan; ¹⁸Institute of Hematology and Blood Diseases Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Tianjin, China; ¹⁹Department of Hematology, Medical University of Lodz, Lodz, Poland; ²⁰Department of Hematology, Peter MacCallum Cancer Centre, The Royal Melbourne Hospital and The University of Melbourne, Melbourne, Australia; and ²¹Department of Hematology, Erasmus University Medical Center Rotterdam and Erasmus MC Cancer Institute, Rotterdam, The Netherlands

The European LeukemiaNet (ELN) genetic risk classifications were developed based on data from younger adults receiving intensive chemotherapy. Emerging

analyses from patients receiving less-intensive therapies prompted a proposal for an ELN genetic risk classification specifically for this patient population.

Introduction

European LeukemiaNet (ELN) genetic risk classifications have been used widely in clinical practice and in clinical trials.^{1,2} These classifications were based exclusively on data from patients who received intensive chemotherapy and were not intended for prognostic stratification of older and/or unfit patients receiving less-intensive therapeutic options. Recent attempts to validate the 2017 and 2022 ELN risk classifications in older patients treated with less-intensive regimens have proven suboptimal, with most individuals classified as adverse risk.^{3,4} This supported the need for a new genetic classifier relevant for stratifying prognostic outcomes in patients receiving hypomethylating agent (HMA)-based regimens alone or in combination with either the B-cell leukemia/lymphoma 2 inhibitor venetoclax (VEN) or azacitidine (AZA) with the IDH1 inhibitor ivosidenib (IVO) for IDH1-mutated acute myeloid leukemia (AML).^{5,6}

HMA monotherapy

Regarding the prognostic impact of genetic factors in patients treated with HMA monotherapy, analysis of cytogenetic data

from the phase 3 clinical trial evaluating AZA vs standard of care indicated a 31% to 46% reduced risk of death for patients with adverse-risk karyotype treated with AZA compared with conventional care.⁷ Gene mutation analyses were only available in one-third of patients and were largely inconclusive.

A recently published analysis of 604 patients treated in a randomized clinical trial evaluating gadecebin vs treatment with either AZA, decitabine, or low-dose cytarabine demonstrated that both the 2017 and 2022 ELN risk classifications had suboptimal utility in predicting clinical outcome.³ Remodeling and multivariable analysis identified 3 distinct prognostic risk groups based on the genomic status of 3 genes: a low-risk group identified by *DDX41*^{mut} (germ line in 61% cases); a high-risk group characterized by the presence of either *FLT3*-ITD^{pos} or *TP53*^{mut}, and an intermediate-risk group defined by the absence of *DDX41*^{mut}, *FLT3*-ITD and *TP53*^{mut}.

HMAs plus VEN

There has been a steady increase in the number of studies investigating the role of genomic factors in determining response and survival outcomes after HMA/VEN-based

therapy.^{4,8-16} Similar to the HMA monotherapy study,³ a pooled analysis of 279 patients treated with AZA/VEN in phase 1b and phase 3 trials demonstrated an inability of either the 2017 or 2022 ELN risk classifications to clearly stratify prognostic outcomes.⁴ Reanalysis using a bioinformatic-based approach delineated 3 novel risk groups based on a 4-gene classifier: a higher-benefit group defined by *TP53*^{wt}, *KRAS/NRAS*^{wt}, and *FLT3-ITD*^{neg} (52% patients); an intermediate-benefit group by *FLT3-ITD*^{pos} and/or *KRAS*^{mut} and/or *NRAS*^{mut} (25.4%); and a lower-benefit group defined solely by *TP53*^{mut} (22.6%).⁴ The median overall survival (OS) times for the 3 risk groups were 26.5 months, 12.1 months, and 5.5 months, respectively. The adverse prognostic impact of *TP53*^{mut} or *FLT3-ITD*^{pos} after treatment with AZA/VEN and failure to improve survival above AZA alone has been reported previously.^{9,10} Importantly, an external validation of this 4-gene predictive signature was provided recently by a single-center study of 159 patients treated with HMA/VEN.¹⁵

Within these newly defined higher- and intermediate-benefit groups, the most frequently occurring International Consensus Classification disease entities were AML with myelodysplasia-related (MR) gene mutations (corresponding to AML, MR in the World Health Organization fifth edition classification¹⁷), followed by AML with mutated *NPM1*, and AML not otherwise specified.⁴ In patients with *NPM1*- or *IDH2*-mutated AML, exploratory analyses identified a negative prognostic impact in the setting of concomitant signaling gene mutations (*FLT3-ITD*^{pos}, *KRAS*^{mut}, and *NRAS*^{mut}), which is consistent with their biological associations as mediators of VEN resistance. For example, *NPM1*^{mut} AML without signaling gene mutations had a median OS of 39 months compared with only 9.9 months in the presence of a co-occurring *FLT3-ITD*^{pos}, *KRAS*^{mut}, and/or *NRAS*^{mut}. Similar trends were found for *IDH2*^{mut} (median OS, 36.9 vs 12.2 months), *RUNX1*^{mut} (median OS, 32.5 vs 9.3 months), and AML with MR gene mutations (median OS, 22.9 vs 12.9 months), if an activated kinase pathogenic variant was absent or present, respectively.

DDX41 mutations identify a particularly favorable-risk group among patients treatment with HMA therapy³ and similarly appear to demonstrate a favorable prognosis in patients treated with AZA/VEN, with 2 recent studies reporting a high response rate and favorable outcome for patients with *DDX41*^{mut} AML.^{13,14} At 2 years, *DDX41*-mutated AML was associated with an OS probability of 60.1% (median OS, 27.8 months) after HMA monotherapy³ and 91.1% (median OS, not reached) after HMA plus VEN-based therapy.¹³

AZA and IVO

A randomized phase 3 study established AZA/IVO as a new treatment standard for patients with *IDH1*^{mut} AML, with improved OS for AZA/IVO, compared with AZA/PBO (24.0 vs 7.9 months).⁶ With additional follow-up, the median OS in the AZA/IVO arm has extended to 29.3 months.¹⁸ Although this compares favorably with 10.2 months for *IDH1*^{mut} patients treated with AZA/VEN,¹⁹ such uncontrolled cross-trial comparisons should be interpreted cautiously. Genes frequently comutated with *IDH1* include *DNMT3A*, *SRSF2*, and *RUNX1*.²⁰ Ad hoc subgroup analyses suggest that AZA/IVO has favorable outcome in *IDH1*^{mut} AML even if co-occurring with either these or

receptor tyrosine kinase mutations.²⁰ Therefore, based on the available data, it is appropriate to categorize all patients with *IDH1*^{mut} AML as favorable in the context of AZA/IVO therapy.

Genetic risk classification

Table 1 outlines a proposed ELN genetic risk classification framework for patients with newly diagnosed AML receiving less-intensive HMA-based therapies (2024 ELN Less-Intensive). The 2024 ELN Less-Intensive genetic risk classification is applicable to patients receiving HMA monotherapy, HMA/VEN, or AZA/IVO (for *IDH1*^{mut} AML). For patients with favorable-risk disease, the median OS times have been reported to be >24 months. For adverse-risk patients, OS ranges between 5 to 8 months, whereas the remaining patients are categorized as intermediate risk. Table 2 gives an overview of reported survival times for selected genetic subgroups.

Although patients with *NPM1*^{mut} or *IDH2*^{mut} AML appear sensitive to VEN treatment,⁸ favorable clinical outcome appears limited to cases lacking activated signaling gene comutations. *DDX41*^{mut} (two-thirds of germ line origin) has increasingly been linked to favorable outcomes in AML, after both HMA monotherapy or VEN-based combination therapies.^{3,13} Finally, all patients with *IDH1*^{mut} have favorable outcomes after AZA/IVO.

TP53^{mut} are universally associated with adverse clinical risk, with poor outcomes for HMA therapy alone or in combination with VEN.^{3,4,10,14-16} Complex karyotype is present in the majority (80%-90%) of *TP53*^{mut} cases. Although the median OS for cytogenetic adverse-risk AML without concurrent *TP53*^{mut} treated with AZA/VEN is 23.4 months,¹⁰ additional confirmatory studies in the setting of less-intensive therapies are needed.

AML with MR mutations as defined by the International Consensus Classification has established specificity for

Table 1. ELN risk classification for patients receiving less-intensive therapies (ELN 2024 Less-Intensive)

Risk category	Genetic abnormality
Favorable	Mutated <i>NPM1</i> (<i>FLT3-ITD</i> ^{neg} , <i>NRAS</i> ^{wt} , <i>KRAS</i> ^{wt} , <i>TP53</i> ^{wt}) Mutated <i>IDH2</i> (<i>FLT3-ITD</i> ^{neg} , <i>NRAS</i> ^{wt} , <i>KRAS</i> ^{wt} , <i>TP53</i> ^{wt}) Mutated <i>IDH1</i> * (<i>TP53</i> ^{wt}) Mutated <i>DDX41</i> † Other cytogenetic and/or molecular abnormalities‡ (<i>FLT3-ITD</i> ^{neg} , <i>NRAS</i> ^{wt} , <i>KRAS</i> ^{wt} , <i>TP53</i> ^{wt})
Intermediate	Other cytogenetic and molecular abnormalities‡ (<i>FLT3-ITD</i> ^{pos} and/or <i>NRAS</i> ^{mut} and/or <i>KRAS</i> ^{mut} , <i>TP53</i> ^{wt})
Adverse	Mutated <i>TP53</i>

This classification does not apply to patients who have received prior treatment with an HMA.

*Favorable risk applies specifically to patients treated with AZA + IVO, irrespective of the presence of activating signaling gene mutations.

†Identification of a *DDX41* mutation at near-heterozygous frequency should prompt consideration of germ line *DDX41* mutation.

‡For many cytogenetic and molecular abnormalities, single or as coaberrations, no data are currently available; they are tentatively categorized as favorable and intermediate-risk depending on the absence or presence of activating signaling gene mutations.

Table 2. Overview on median OS times by genetic marker

Genetic marker	Median OS, mo	Reference
Favorable-risk group		
Mutated <i>NPM1</i> (<i>FLT3-ITD</i> ^{neg} , <i>NRAS</i> ^{wt} , <i>KRAS</i> ^{wt} , <i>TP53</i> ^{wt})	39	4
Mutated <i>IDH2</i> (<i>FLT3-ITD</i> ^{neg} , <i>NRAS</i> ^{wt} , <i>KRAS</i> ^{wt} , <i>TP53</i> ^{wt})	37	4
Mutated <i>IDH1</i> * (<i>TP53</i> ^{wt})	29	6,17
Mutated <i>DDX41</i>	>24	3,13
AML with MR gene mutations (<i>FLT3-ITD</i> ^{neg} , <i>NRAS</i> ^{wt} , <i>KRAS</i> ^{wt} , <i>TP53</i> ^{wt})	23	4
Intermediate-risk group		
AML with MR gene mutations (<i>FLT3-ITD</i> ^{pos} and/or <i>NRAS</i> ^{mut} and/or <i>KRAS</i> ^{mut} ; <i>TP53</i> ^{wt})	13	4
Other cytogenetic and molecular abnormalities (<i>FLT3-ITD</i> ^{pos} and/or <i>NRAS</i> ^{mut} and/or <i>KRAS</i> ^{mut} ; <i>TP53</i> ^{wt})	12	4
Adverse-risk group		
Mutated <i>TP53</i>	5-8	3,4,7,10,14-16

*Favorable risk (median OS time) applies specifically to patients treated with AZA + IVO, irrespective of the presence of activating signaling gene mutations.

secondary AML.²¹ A recent study indicates that patients with AML and MR mutations are particularly responsive to HMA/ VEN²²; consistent with other reports linking splicing factor variants or *ASXL1* mutations to improved outcomes after HMA/ VEN.^{23,24} However, even within this category, the copresence of signaling gene mutations is associated with inferior prognosis.⁴

This iteration of the 2024 ELN Less-Intensive AML classification was based on currently available genetic data from less-intensively treated patients, which still remain limited. For AML subgroups with specific gene rearrangements (eg, translocations and inversions), characterization of their impact on prognosis was not possible, owing to their rarity in older age patients, resulting in a paucity of available data. The current classification also has limited relevance to patients with prior myeloproliferative neoplasm or prior exposure to HMA therapies, including AML after an antecedent myelodysplastic syndrome, because such patients were generally excluded from clinical trials of VEN in AML. In terms of low-dose cytarabine/ VEN-based treatment and genomic determinants of outcome, we have not included specific comments in the current ELN Less-Intensive classification, because these analyses have yet to be published.

The 2022 ELN recommendations additionally emphasize the role of measurable residual disease (MRD) assessment for comprehensive risk assessment, which has become integral to the management of patients treated with intensive chemotherapy. In patients treated with HMA-based therapies, MRD data are just emerging, and they suggest that MRD monitoring also plays an important and complementary role in informing prognosis.^{25,26}

Acknowledging its limitations, this ELN genetic risk classification provides a basis for consensus stratification of patients receiving less-intensive therapies. It will be important for these 2024 ELN Less-Intensive recommendations to be validated in future clinical trials, as well as large real-world data sets. With the treatment landscape in older/unfit patients continuing to evolve (eg, with addition of FLT3 and menin inhibitors), it is

expected that this classification will continue to undergo iterative future refinement.

Acknowledgments

The authors acknowledge Rüdiger Hehlmann for his continuous generous support of these recommendations on behalf of the European LeukemiaNet.

H. Döhner is supported by Sonderforschungsbereich 1074 "Experimental models and clinical translation in leukemia" and Forschungsgruppe 2674, project A02 "Genetic landscape of acute myeloid leukemia in older patients," both funded by the Deutsche Forschungsgemeinschaft. A.H.W. is supported by the Australian National Health and Medical Research Council, Victorian Cancer Agency, Metcalf Family Fellowship, and the Medical Research Future Fund. C.D.D. is supported by the Leukemia & Lymphoma Society Scholar in Clinical Research Award.

Authorship

Contribution: H. Döhner, C.D.D., A.H.W., and B.L. wrote the draft version of the manuscript; and all authors reviewed and approved the final version of the manuscript.

Conflict-of-interest disclosure: H. Döhner reports advisory role with honoraria for AbbVie, AstraZeneca, Gilead, Janssen, Jazz, Pfizer, Servier, Stemline Therapeutics, and Syndax and clinical research funding (to institution) from AbbVie, Astellas, Bristol Myers Squibb (BMS), Celgene, Jazz Pharmaceuticals, Kronos Bio, and Servier. C.D.D. reports honoraria/consulting fees from AbbVie, Agios/Servier, Astellas, Celgene/BMS, Cleave, Foghorn, Genentech, GenMab, GSK, Novartis, Notable Labs, and Takeda and research grants (to the institution) from AbbVie, Agios/Servier, Astex, Calithera, Celgene/BMS, Cleave, Foghorn, Immune-Onc Therapeutics, and Loxo. A.H.W. reports advisory role with honoraria for AbbVie, Agios, Amgen, Astellas, AstraZeneca, Roche, BMS, Celgene, Gilead, Pfizer, Janssen, Jazz, Novartis, and Servier; clinical research funding (to institution) from AbbVie, Servier, Celgene/BMS, AstraZeneca, Amgen, and Novartis; and receives a fraction of royalty payments from the Walter and Eliza Hall Institute of Medical Research related to venetoclax. B.L. reports advisory role with honoraria for AbbVie, Astellas, BMS, Stemline Pharmaceuticals/Menarini, Syndax Pharmaceuticals, Ryvu Therapeutics, Wugen Inc, and CureVac and royalties from UpToDate. F.R.A. reports consulting for Jasper Biote. C.C. reports advisory role with honoraria for AbbVie, Amgen, Astellas, AstraZeneca, Berlin-Chemie, BMS, Celgene, Daiichi Sankyo, Eurocept Pharmaceuticals, Gilead, Janssen, Jazz, and Novartis and clinical research funding from AbbVie, BMS, Celgene, and Jazz Pharmaceuticals. H. Dombret reports honoraria/consulting fees from AbbVie, Amgen, Astellas, Celgene/BMS,

Daiichi Sankyo, Incyte, Jazz Pharmaceuticals, Pfizer, and Servier and research funding from Amgen, Astellas, Celgene/BMS, Incyte, Jazz Pharmaceuticals, and Pfizer. B.L.E. has received research funding from Novartis and Calico; has received consulting fees from AbbVie; and is a member of the scientific advisory board and shareholder for Neomorph Inc, TenSixteen Bio, Skyhawk Therapeutics, and Exo Therapeutics. P.F. reports honoraria and, as French Myelodysplastic Syndrome group chairperson, research support from Celgene/BMS, Novartis, AbbVie, Jazz, Janssen, and Agios. R.A.L. reports receiving consulting or advisory fees from AbbVie, Amgen, Ariad/Takeda, Astellas, Celgene/BMS, Curis, CVS Caremark, Epizyme, Function Oncology, ImmunoGen, Jazz Pharmaceuticals, Kling Biotherapeutics, Medpace, MorphoSys, Novartis, and Servier; clinical research support to his institution from Astellas, Biomea, Celgene, Collectis, Daiichi Sankyo, Forty Seven/Gilead, Novartis, and Rafael Pharmaceuticals; and royalties from UpToDate. R.L.L. reports supervisory board fees from Qiagen; scientific advisor fees from Mission Bio, Zentalis Pharmaceuticals, Ajax, Auron, Prelude, and C4 Therapeutics, for which he receives equity; research support from Calico, Zentalis Pharmaceuticals, and Ajax; consultation fees from Incyte and Janssen; and honoraria from AstraZeneca for invited lectures. Y.M. reports honoraria from Nippon Shinyaku, BMS, Novartis, Sumitomo Pharma, Kyowa Kirin, AbbVie, Daiichi Sankyo, Takeda, Janssen Pharmaceutical, Astellas, Pfizer, Eisai, Otsuka Pharmaceutical, and Sumitomo Dainippon and research funding from Chugai. G.O. reports advisory role with honoraria for AbbVie, Astellas, BMS, Celgene, Gilead, Servier, Jazz, and Novartis. C.R. reports advisory role with honoraria for AbbVie, Amgen, Astellas, BMS, Celgene, Jazz, Novartis, Pfizer, and Servier and clinical research funding from AbbVie, Novartis, and Pfizer. J.S. reports advisory role with honoraria for AbbVie, Astellas, Jazz Pharmaceuticals, and BMS and clinical research funding from AbbVie, Astellas, Jazz Pharmaceuticals, and Jose Carreras International Leukemia Foundation. E.M.S. reports advisory board fees from Novartis, Pinotbio, Janssen, BMS, Agios, Jazz, Menarini, Genentech, Genesis, AbbVie, Neoleukin, Gilead, Syndax, OnCusp Therapeutics, CTI BioPharma, Foghorn, Servier, Calithera, Daiichi Sankyo, Aptose, Syros, Astellas, Ono Pharma, and Blueprint;

honoraria from Kura; safety monitoring roles with Epizyme and Collectis; research funding from Eisai and BMS; and equity from Auron. M.S.T. reports advisory board fees from AbbVie, Daiichi Sankyo, Molecular, Orsenix, KAHN, Oncolyze, Jazz Pharma, Roche, BioSight, Novartis, Innate Pharmaceuticals, Kura, Syros Pharmaceuticals, Ipsen Pharmaceuticals, and Cellularity; research funding from AbbVie, Orsenix, BioSight, GlycoMimetics, Rafael Pharmaceuticals, and Amgen; royalties from UpToDate; and serving as Chair of Differentiation Adjudication Committee at Foghorn (Foghorn259) and chair of data safety monitoring board for HOVON150. H.-F.T. reports advisory role with honoraria for AbbVie, Alexion, Celgene, Daiichi Sankyo, Novartis, and Roche and research funding from Celgene. J.W. reports advisory role with honoraria for AbbVie. A.W. reports advisory role with honoraria for AbbVie, Astellas, BMS, Celgene, Gilead, Janssen, Jazz Pharmaceuticals, Novartis, Pfizer, and Servier and clinical research funding from Jazz Pharmaceuticals. The remaining authors declare no competing financial interests.

ORCID profiles: H.D., 0000-0003-2116-5536; C.D.D., 0000-0001-9003-0390; C.C., 0000-0001-5041-6678; B.L.E., 0000-0003-0197-5451; R.A.L., 0000-0001-9168-3203; D.N., 0000-0002-4737-1103; C.R., 0000-0002-3791-0548; J.S., 0000-0002-7966-0356; J.W., 0000-0001-9437-9151; A.W., 0000-0001-7909-457X; A.H.W., 0000-0002-7514-3298; B.L., 0000-0001-8982-5217.

Correspondence: Hartmut Döhner, Department of Internal Medicine III, Ulm University Hospital, Albert-Einstein-Allee 23, Ulm 89081, Germany; email: hartmut.doehner@uniklinik-ulm.de.

Footnote

Submitted 15 May 2024; accepted 12 July 2024; prepublished online on *Blood* First Edition 12 August 2024. <https://doi.org/10.1182/blood.2024025409>.

REFERENCES

- Döhner H, Estey E, Grimwade D, et al. Diagnosis and management of acute myeloid leukemia in adults: 2017 ELN recommendations from an international expert panel. *Blood*. 2017;129(4):424-447.
- Döhner H, Wei AH, Appelbaum FR, et al. Diagnosis and management of acute myeloid leukemia in adults: 2022 recommendations from an international expert panel, on behalf of the European LeukemiaNet. *Blood*. 2022;140(12):1345-1377.
- Jahn E, Saadati M, Fenaux P, et al. Clinical impact of the genomic landscape and leukemogenic trajectories in non-intensively treated elderly acute myeloid leukemia patients. *Leukemia*. 2023;37(11):2187-2196.
- Döhner H, Pratz KW, DiNardo CD, et al. Genetic risk stratification and outcomes among treatment-naïve patients with AML treated with venetoclax and azacitidine. *Blood*. 2024;144(21):2211-2222.
- DiNardo CD, Jonas BA, Pullarkat V, et al. Venetoclax and azacitidine combination for treatment-naïve patients with AML. *N Engl J Med*. 2020;383(7):617-629.
- Montesinos P, Recher C, Vives S, et al. Ivosidenib and azacitidine in IDH1-mutated acute myeloid leukemia. *N Engl J Med*. 2022;386(16):1519-1531.
- Döhner H, Dolnik A, Tang L, et al. Cytogenetics and gene mutations influence survival in older patients with acute myeloid leukemia treated with azacitidine or conventional care. *Leukemia*. 2018;32(12):2546-2557.
- DiNardo CD, Tiong IS, Quaglieri A, et al. Molecular patterns of response and treatment failure after frontline venetoclax combinations in older patients with AML. *Blood*. 2020;135(11):791-803.
- Konopleva M, Thirman MJ, Pratz KW, et al. Impact of FLT3 mutation on outcomes after venetoclax and azacitidine for patients with treatment-naïve acute myeloid leukemia. *Clin Cancer Res*. 2022;28(13):2744-2752.
- Pollyea DA, Pratz KW, Wei AH, et al. Outcomes in patients with poor-risk cytogenetics with or without TP53 mutations treated with venetoclax and azacitidine. *Clin Cancer Res*. 2022;28(24):5272-5279.
- Pollyea DA, DiNardo CD, Arellano ML, et al. Impact of venetoclax and azacitidine in treatment-naïve patients with acute myeloid leukemia and IDH1/2 mutations. *Clin Cancer Res*. 2022;28(13):2753-2761.
- Garciaz S, Hospital MA, Alary AS, et al. Azacitidine plus venetoclax for the treatment of relapsed and newly diagnosed acute myeloid leukemia patients. *Cancers (Basel)*. 2022;14(8):2025.
- Bataller A, Loghavi S, Gerstein Y, et al. Characteristics and clinical outcomes of patients with myeloid malignancies and DDX41 variants. *Am J Hematol*. 2023;98(11):1780-1790.
- Gangat N, Karrar O, Ifthikhar M, et al. Venetoclax and hypomethylating agent combination therapy in newly diagnosed acute myeloid leukemia: genotype signatures for response and survival among 301 consecutive patients. *Am J Hematol*. 2024;99(2):193-202.
- Bataller A, Bazinet A, DiNardo CD, et al. Prognostic risk signature in patients with acute myeloid leukemia treated with hypomethylating agents and venetoclax. *Blood Adv*. 2024;8(4):927-935.
- Miyashita N, Onozawa M, Matsukawa T, et al. Novel stratification for newly diagnosed acute myeloid leukaemia treated with venetoclax-based therapy in the real world: Hokkaido Leukemia Net Study. *Br J Haematol*. 2024;204(4):1549-1553.
- Khoury JD, Solary E, Abal O, et al. The 5th edition of the World Health Organization classification of haematolymphoid tumours: myeloid and histiocytic/dendritic neoplasms. *Leukemia*. 2022;36(7):1703-1719.
- De Botton S, Montesinos P, Vives Polo S, et al. Updated efficacy and safety data from the AGILE study in patients with newly diagnosed acute myeloid leukemia treated with ivosidenib + azacitidine compared to placebo + azacitidine. *J Clin Oncol*. 2023;41(suppl 16):7012.

19. Pratz KW, Jonas BA, Pullarkat V, et al. Long-term follow-up of VIALE-A: venetoclax and azacitidine in chemotherapy-ineligible untreated acute myeloid leukemia. *Am J Hematol*. 2024;99(4):615-624.
20. De Botton S, Choe S, Marchione DM, et al. Molecular characterization of clinical response in patients with newly diagnosed acute myeloid leukemia treated with ivosidenib + azacitidine compared to placebo + azacitidine. *JCO*. 2022;40(suppl 16):7019.
21. Lindsley RC, Mar BG, Mazzola E, et al. Acute myeloid leukemia ontogeny is defined by distinct somatic mutations. *Blood*. 2015; 125(9):1367-1376.
22. Shimony S, Garcia JS, Keating J, et al. Molecular ontogeny underlies the benefit of adding venetoclax to hypomethylating agents in newly diagnosed AML patients. *Leukemia*. 2024;38(7):1494-1500.
23. Senapati J, Urrutia S, Loghavi S, et al. Venetoclax abrogates the prognostic impact of splicing factor gene mutations in newly diagnosed acute myeloid leukemia. *Blood*. 2023;142(19):1647-1657.
24. Rahmani NE, Ramachandra N, Sahu S, et al. ASXL1 mutations are associated with distinct epigenomic alterations that lead to sensitivity to venetoclax and azacytidine. *Blood Cancer J*. 2021;11(9):157.
25. Pratz KW, Jonas BA, Pullarkat V, et al. Measurable residual disease response and prognosis in treatment-naïve acute myeloid leukemia with venetoclax and azacitidine. *J Clin Oncol*. 2022;40(8):855-865.
26. Othman J, Tiong IS, O'Nions J, et al. Molecular MRD is strongly prognostic in patients with NPM1-mutated AML receiving venetoclax-based nonintensive therapy. *Blood*. 2024;143(4):336-341.

© 2024 American Society of Hematology. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.